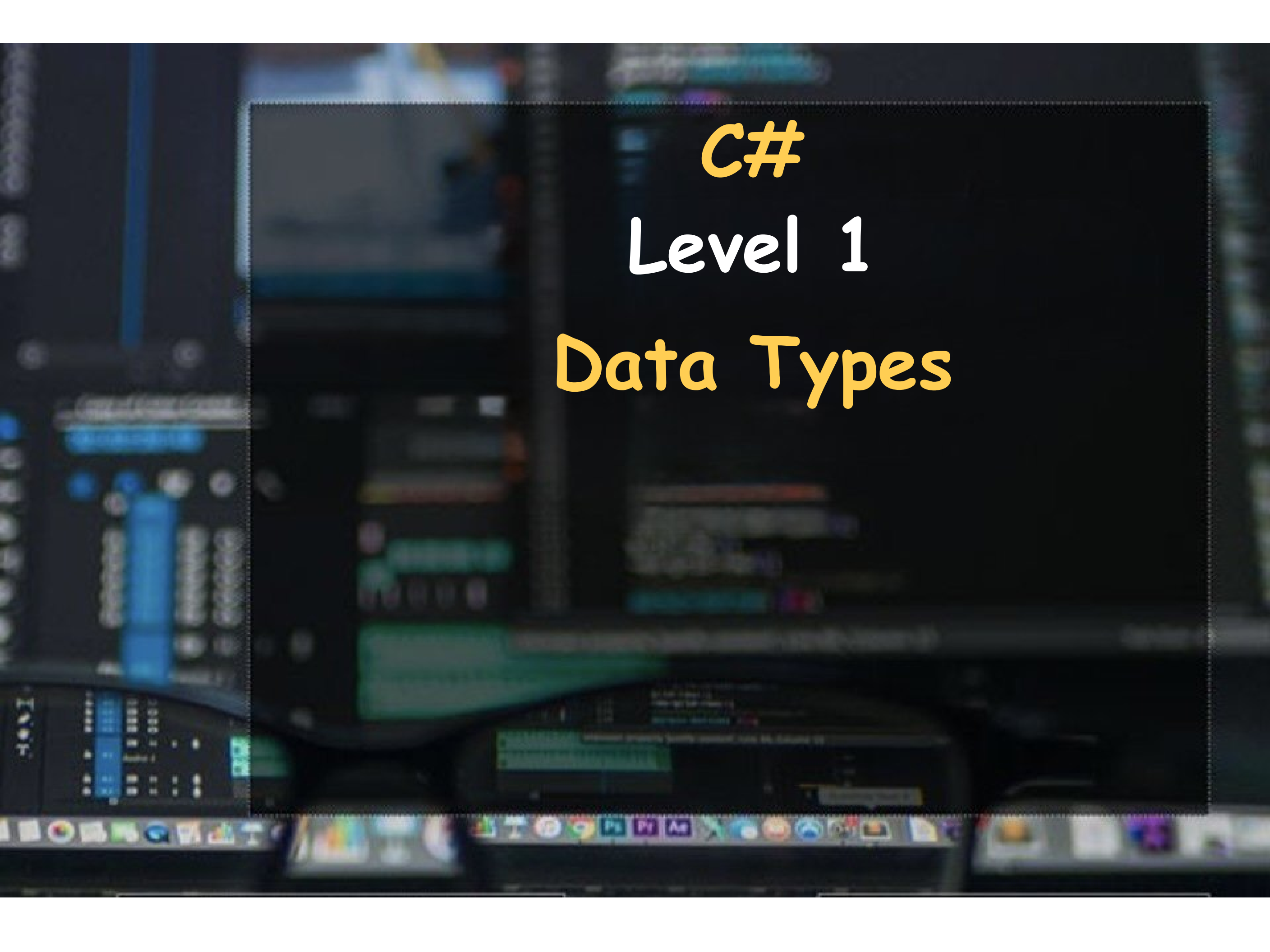
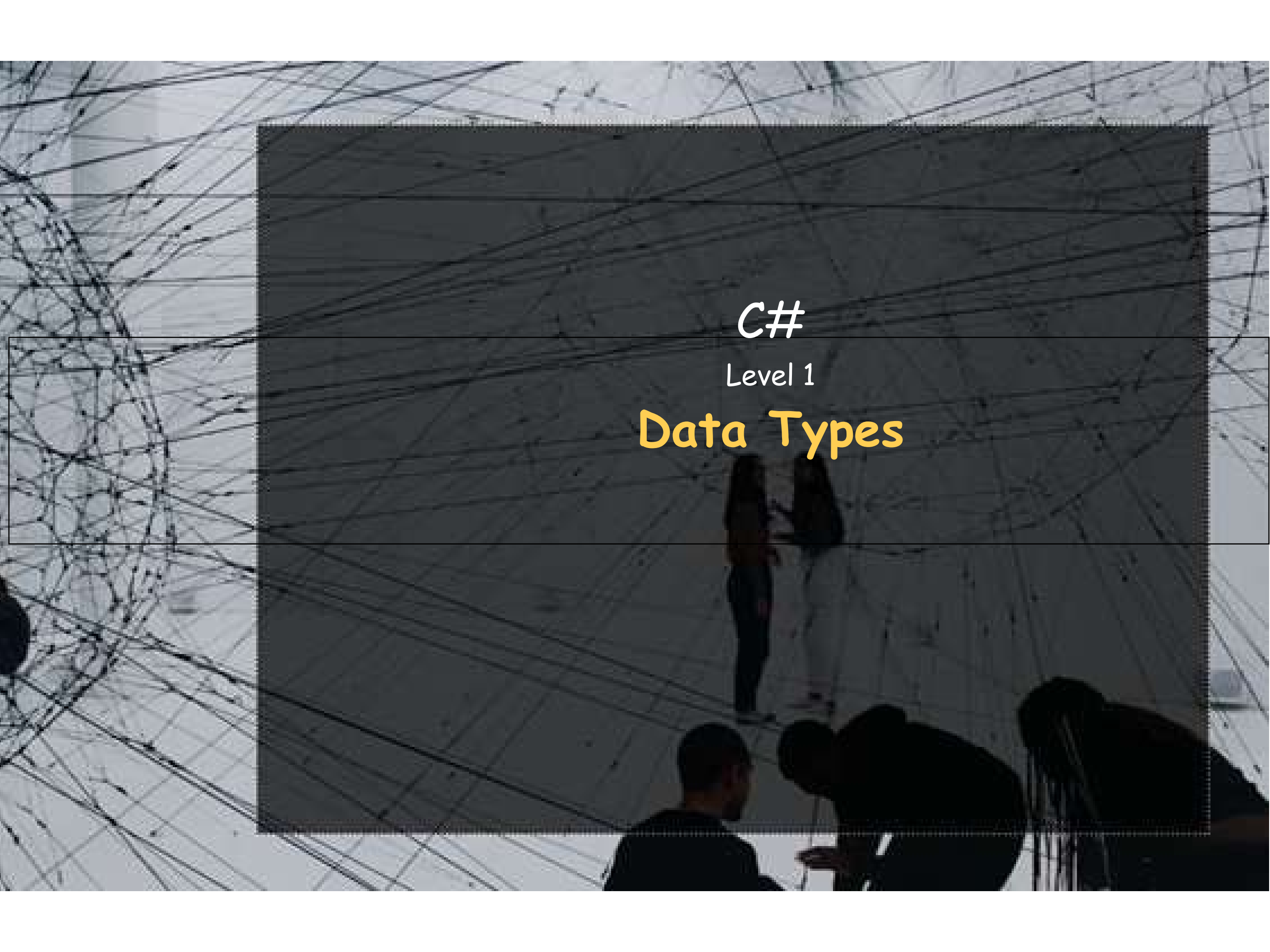




C#

Level 1

Data Types



C#

Level 1

Data Types

Definition

Data Type Classification

-  Data types define what kind of value a variable can store.
-  C# is strongly typed, so every variable has a specific type.

Explanation

Value Types

-  Store the actual value directly.

-  Common types include ``int``, ``long``, ``float``, ``double``, ``decimal``, ``char``, and ``bool``.

Reference Types

-  Store a reference to an object rather than the object data directly.

-  Common types include ``string``, ``object``, arrays, classes, and interfaces.

Basic Data Types

-  ``int`` → whole numbers.

-  ``long`` → large whole numbers.

-  ``float`` → decimal numbers using the ``f`` suffix.

-  ``double`` → decimal numbers with higher precision than ``float``.

-  ``decimal`` → high-precision decimal values, commonly used for financial calculations.

-  ``char`` → one character using single quotes.

-  ``bool`` → ``true`` or ``false``.

-  ``string`` → text using double quotes.

Procedures

1. Declare a variable with a suitable data type.
2. Assign a value to the variable.
3. Use the variable in the program.
4. Display or process the stored value when needed.

💡 Example

```
string name = "Sara";  
int age = 25;  
double height = 1.65;  
char grade = 'A';  
bool isStudent = true;  
Console.WriteLine(name);  
Console.WriteLine(age);  
Console.WriteLine(height);  
Console.WriteLine(grade);  
Console.WriteLine(isStudent);
```

➡ Result

```
Sara  
25  
1.65  
A  
True
```

Note

- ◆ ``string`` can contain multiple characters, while ``char`` stores only one.
- ◆ ``var`` is not a separate data type; the compiler determines the variable's type automatically.

⊘ Warning

◆ `char` requires single quotes:

```
char letter = 'A';    // ✓
```

◆ `string` requires double quotes:

```
string name = "Sara"; // ✓
```

◆ `float` requires the `f` suffix:

```
float x = 10.5f;      // ✓
```



Comparison

▨ Value Types

- ✓ Store the value directly.
- ✓ Include ``int``, ``double``, ``char``, ``bool``, etc.
- ✓ Suitable for individual values such as numbers and characters.

▤ Reference Types

- ✓ Store a reference to an object.
- ✓ Include ``string``, arrays, classes, and ``object``.
- ✓ Suitable for objects and collections of data.

✦ Summary

- ✓ Data types determine what kind of data a variable can store.
- ✓ Value types store values directly.
- ✓ Reference types store references to objects.
- ✓ ``int`` → whole numbers.
- ✓ ``double`` → decimal numbers.
- ✓ ``char`` → one character.
- ✓ ``bool`` → true / false.
- ✓ ``string`` → text.
- ✓ ``var`` → compiler determines the type automatically.

Thank you 😊